

# The Ambient-Realization Gate Between Reset Packets and Riesz Clouds

A Type-Correct Primal–Adjoint Coupling Theorem

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## Abstract

RH-161 reduced the first determinant gate to a packet-to-Riesz estimate, but its abstract theorem places the reset projection and transfer operator on one Hilbert space. In the archived physical construction the reset packet is a spectral subspace of a source-memory Gram matrix, while the moving cloud is a Riesz subspace of a noisy two-step transfer operator. Equal rank does not identify these spaces.

We make the missing ambient realization explicit. Let  $J : \mathcal{E} \rightarrow \mathcal{H}$  be an isometry, let  $P$  reduce a source operator  $M$ , and put  $\hat{P} = JPJ^*$ . For a transfer operator  $A$  define the primal and adjoint defects

$$D = AJ - JM, \quad D^\sharp = A^*J - JM^*.$$

We prove the directed bounds

$$\|(I - \hat{P})A\hat{P}\| \leq \|DP\|, \quad \|\hat{P}A(I - \hat{P})\| \leq \|D^\sharp P\|.$$

Thus a type-correct realization supplies exactly the two feedback couplings needed by a Schur resolvent argument. A two-dimensional witness proves that without  $J$  the source packet gives no coupling information. We also record the polar repair and its commutator debt for a near-isometric map. A 512-case finite audit has zero bound failures. No canonical prime-dynamics  $J$  or all-level defect estimate is constructed, so physical interface R and Gate A remain open.

## 1 Why the ambient space is part of the theorem

The RH-160 reset construction diagonalizes positive memory Grams on a source or clock space [Program, 2026]. RH-80 and RH-161 require a reducing Riesz cloud for a transfer operator. Writing both outputs as a rank- $r$  projection suppresses the domain and is therefore not type-correct. A rank, a list of eigenvalues, or a fitted frame does not define a functorial map between the two spaces.

Let  $\mathcal{E}$  be the source-memory Hilbert space and  $\mathcal{H}$  the transfer space. Let  $M \in \mathcal{B}(\mathcal{E})$ ,  $A \in \mathcal{B}(\mathcal{H})$ , and let  $P$  be a finite-rank orthogonal projection satisfying  $PM = MP$ . An *ambient realization* is an isometry  $J : \mathcal{E} \rightarrow \mathcal{H}$ . Its realized packet is

$$\hat{P} = JPJ^*, \quad \hat{Q} = I - \hat{P}.$$

The complement includes both  $J(\text{ran}(I - P))$  and any directions outside  $J\mathcal{E}$ ; this is essential when the dimensions differ.

## 2 The primal–adjoint realization theorem

**Theorem 1** (Realization defects control both couplings). *With the notation above, define*

$$D = AJ - JM, \quad D^\sharp = A^*J - JM^*.$$

*Then*

$$\|\widehat{Q}A\widehat{P}\| \leq \|DP\|, \tag{1}$$

$$\|\widehat{P}A\widehat{Q}\| \leq \|D^\sharp P\|. \tag{2}$$

*Consequently the two-way feedback product obeys*

$$\|\widehat{P}A\widehat{Q}\| \|\widehat{Q}A\widehat{P}\| \leq \|D^\sharp P\| \|DP\|.$$

*Proof.* For  $x \in \text{ran } P$ , reduction gives  $MPx \in \text{ran } P$ , and hence  $JMPx \in \text{ran } \widehat{P}$ . Therefore

$$\widehat{Q}AJx = \widehat{Q}(JMx + Dx) = \widehat{Q}Dx.$$

Since  $J$  is isometric on  $\text{ran } P$ , taking the supremum proves (1). Applying the same argument to  $A^*$  and  $M^*$  gives  $\|\widehat{Q}A^*\widehat{P}\| \leq \|D^\sharp P\|$ . Taking adjoints proves (2). Multiplication gives the final inequality.  $\square$

The adjoint defect is not a cosmetic duplication. Nonnormal dynamics can have a small right residual and a large left residual. Later Schur bounds use their product, while directional graph bounds distinguish them.

**Corollary 2** (Exact intertwining). *If  $AJ = JM$  and  $A^*J = JM^*$  on  $\text{ran } P$ , then  $\widehat{P}$  reduces  $A$ . Any contour separating the realized packet spectrum from its complement already has  $\widehat{P}$  as its Riesz projection.*

*Proof.* Both off-diagonal blocks vanish by Theorem 1.  $\square$

## 3 No coupling bound without a realization

**Proposition 3** (Two-dimensional non-identifiability witness). *Fix the one-dimensional source data  $\mathcal{E} = \mathbb{C}$ ,  $M = 0$ , and  $P = I$ . Let  $A = \text{diag}(0, 1)$  on  $\mathbb{C}^2$ . There are isometries  $J_0, J_1 : \mathcal{E} \rightarrow \mathbb{C}^2$  realizing the identical source packet for which*

$$\|(I - J_0 J_0^*) A J_0 J_0^*\| = 0, \text{quad} \|(I - J_1 J_1^*) A J_1 J_1^*\| = \frac{1}{2}.$$

*Proof.* Take  $J_0(1) = e_1$  and  $J_1(1) = 2^{-1/2}(e_1 + e_2)$ . The first range is an eigenspace. In the orthonormal basis  $2^{-1/2}(e_1 + e_2), 2^{-1/2}(e_1 - e_2)$ , the off-diagonal entry of  $A$  has absolute value  $1/2$ .  $\square$

Thus packet positivity, overlap floors, and source-space rank do not bound a transfer-space coupling. The map  $J$  and its dynamical defects are genuine physical inputs, not coordinate notation.

## 4 Near-isometries and polar repair

Suppose a proposed map  $J_0$  obeys

$$\|J_0^* J_0 - I\| \leq \eta < 1.$$

Then  $G = J_0^* J_0$  is positive invertible and  $J = J_0 G^{-1/2}$  is an isometry. Functional calculus gives

$$\|G^{-1/2}\| \leq (1 - \eta)^{-1/2}, \quad \|G^{-1/2} - I\| \leq (1 - \eta)^{-1/2} - 1.$$

However,

$$\begin{aligned} AJ - JM &= (AJ_0 - J_0 M)G^{-1/2} \\ &\quad + J_0(MG^{-1/2} - G^{-1/2}M). \end{aligned}$$

The second term vanishes only when the source dynamics commutes with the Gram correction. A numerical orthogonalization that omits this commutator does not certify the physical intertwining defect [Stewart and Guang Sun, 1990].

## 5 Gauge covariance and a canonicity test

An ambient realization should depend on the source subspace, not on a chosen basis. The following elementary identity is useful in implementation.

**Proposition 4** (Source-gauge covariance). *Let  $U$  be unitary on  $\mathcal{E}$  and transform the source data by*

$$M' = U M U^*, \quad P' = U P U^*, \quad J' = J U^*.$$

*Then the realized packet and directed defect norms are unchanged:*

$$J' P' J'^* = J P J^*,$$

*and*

$$\|(AJ' - J'M')P'\| = \|(AJ - JM)P\|,$$

*with the analogous identity for the adjoint defect.*

*Proof.* The first identity follows by cancellation of adjacent  $U^*U$  factors. Also

$$(AJ' - J'M')P' = (AJ - JM)PU^*,$$

and right multiplication by a unitary preserves the norm. Apply the same calculation to  $A^*$  and  $M^*$ .  $\square$

This gives a direct falsifier for a proposed physical construction: if a rotation of the reset frame changes the realized transfer packet, the map is frame-fitted rather than packet-canonical. Three further tests remain independent:  $J$  must not use target zero data; its normalization must remain controlled as the dimensions grow; and its common-coordinate transports must be compatible with refinement. Gauge covariance is necessary but not sufficient for those stronger requirements.

## 6 Finite audit and physical frontier

The companion audit samples 512 random triples  $(A, M, J)$  with dimensions 10, 7, and packet rank 3. It evaluates both sides of (1)–(2); no failure occurs. This checks the implementation, not the prime-dynamics hypotheses.

For the archived route, the new subinterface  $X$  consists of:

1. a target-independent ambient map  $J_j$  from each reset-memory space to the transfer/determinant space;
2. a normalization and polar repair with controlled commutator;
3. primal and adjoint packet-defect bounds;
4. compatibility of these maps across the moving cloud schedule.

Only after  $X$  is supplied can RH-161's contour estimate be applied to the physical reset packet. RH-162 proves the type-correct implication and the impossibility of deleting  $X$  from this architecture. It does not construct  $X$ , a Riesz cloud, a canonical determinant, a scattering completion, a self-adjoint operator, a zeta identity, or the Riemann Hypothesis.

## References

- Prime Dynamics Theory Program. Conditional all-level reset dichotomy, 2026. RH-160 manuscript.
- G. W. Stewart and Ji guang Sun. *Matrix Perturbation Theory*. Academic Press, 1990.